Structural Damage Image Classification

UE24CS352A Machine Learning, Section C, Team 16, Project 31
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Problem statement

After an earthquake, someone has to decide which buildings are damaged. Doing that from photographs, instead of a site visit, is the task in Ho and Troncoso, Structural Damage Image Classification (Stanford CS229, 2018). The label is binary. Undamaged is class 0 and damaged is class 1. A useful model has to beat the majority-class rate, and it has to use the fact that a crack or a collapse is a spatial pattern, not a shuffled list of pixel values.

Dataset

The paper used 5,913 labelled photos from the PEER Hub ImageNet collection (3,186 undamaged, 2,727 damaged), each 224 by 224 RGB. Those files are released only after an email request, so this run uses Concrete Crack Images for Classification (Ozgenel, Mendeley Data, 2019, CC BY 4.0), mirrored on Hugging Face as mohammadnajeeb/concrete_crack_images. It is the same kind of decision on a narrower kind of photo: a close-up of concrete, cracked or intact. The archive is balanced: 12,000 and 12,000 in training, 4,000 and 4,000 in validation, and 4,000 and 4,000 in test. Chance accuracy on the test split is 50%. The paper instead made one 90/10 cut and called the 10% both validation and test. We keep the three published splits, so the test number is not the number we tuned on.

Approach

The paper compared three flat-vector classifiers with convolutional networks. We keep that comparison. K-nearest neighbours uses k = 5. Logistic regression uses an L2 penalty with C = 1. The pixel SVM uses an RBF kernel with C = 1 and gamma = 0.001. Those three models cannot take a 224 by 224 by 3 vector (150,528 numbers) at a practical cost, so each photo is resized to 32 by 32 and mapped with the paper's formula, (pixel / 128) - 1. Training uses 1,500 images per class. Validation and test for these three models use 2,000 per class.

The networks are transfer-learned. MobileNetV2 and InceptionV3 start from ImageNet weights. The original trunk is frozen and a new linear layer with two outputs is trained with cross-entropy, which is the softmax layer the paper added. Torchvision's weights expect ImageNet mean and standard deviation, not the paper's TensorFlow scaling, so the network inputs use that mean and standard deviation. A second MobileNet run unfreezes the last four convolutional blocks, which is the paper's response when a frozen network looks bias-limited. A third run adds horizontal flips and small shifts. A final SVM reads a 768-dimensional vector from InceptionV3 block Mixed_6e (global average pooled). That block stands in for the paper's TensorFlow layer 288: a late-middle representation, not the thousand-way head. Gamma for this SVM is chosen on the validation slice. The largest value in the grid, 0.03, scored highest, and the curve was still rising, so 0.03 is the best value we tried, not a proven peak.

Implementation

python train.py reads the zip archives, fits every model, and writes results/metrics.json plus the figures in results/figures. The demo checkpoint is the model with the highest validation accuracy, which is the Mixed_6e SVM. python demo.py runs that pipeline on a photo: Inception features, then the saved SVM. Neural nets see 3,000 training images per class and the full 8,000-image validation and test splits. The feature SVM sees 1,000, 1,200, and 1,200 per class, so its percentages are not on the same count of photos as the networks. Training used an NVIDIA GeForce RTX 4060 Laptop GPU and Adam (learning rate 0.001 on a frozen trunk, 0.0001 after unfreezing).

Results

Flat-pixel logistic regression is already far above the paper's result. Its training accuracy is 97.27% and its test accuracy is 87.90%. The validation curve rises from 76.33% at 300 images to 82.03% at 1,500 images and then stays near 82%, so more photos of the same 32 by 32 vectors are not what is missing. K-nearest neighbours is the odd one. Test accuracy is 75.12%, but the test confusion counts are 2,000 true undamaged, 0 false damaged, 995 missed cracks, and 1,005 found cracks. Every time it says damaged it is right, and it still misses about half of the cracks, because a thin crack in a new place does not land near the same flattened neighbours.

Model	Train	Validation	Test	Test images
K-nearest neighbours (k=5)	79.73%	68.90%	75.12%	4000
Logistic regression (L2, C=1)	97.27%	81.60%	87.90%	4000
SVM, RBF, C=1, gamma=0.001	94.20%	90.30%	92.97%	4000
MobileNetV2, frozen backbone	99.63%	99.29%	99.45%	8000
MobileNetV2, last blocks unfrozen	99.97%	99.70%	99.70%	8000
MobileNetV2, flip and shift	99.17%	99.26%	99.02%	8000
InceptionV3, frozen backbone	99.18%	98.78%	98.46%	8000
SVM on Inception Mixed_6e (gamma=0.03)	99.75%	99.88%	99.96%	2400

On the full test split, MobileNetV2 with the last blocks unfrozen reaches 99.70% (confusion 3,986, 14, 10, 3,990). Freezing the trunk already reaches 99.45%, so the extra 1,528,642 trainable parameters buy a small gain, not a rescue from chance. Flip-and-shift augmentation lowers test accuracy to 99.03%. InceptionV3, the paper's best model at 83% validation, is 98.46% here and slightly behind MobileNetV2. The feature SVM test accuracy is 99.96% on 2,400 images (1,199, 1, 0, 1,200): one false damaged call and no missed cracks. That slice is smaller than the network test, so the 0.26 point gap over the unfrozen MobileNet does not by itself prove a better model.

Conclusions

A model that can see local structure beats a model that sees a flat pixel vector, which is the paper's main empirical claim, and it holds here. What does not hold is the paper's second claim that classical models sit near 50% to 55%. These crack close-ups are an easier problem than PEER's mix of wide shots, clutter, and mislabels, and logistic regression and the pixel SVM already clear 87% and 93% on the test slice. The run does not show that InceptionV3 is the right network for structural damage in general, and it does not show that the same accuracies would survive photographs of collapsed buildings. The supported claim is narrower: on balanced concrete close-ups, with a held-out test split, transfer learning and a mid-network SVM are above 98%, and flattened pixels leave a gap that more of the same training images do not close.